

Gate 6A - Cloud + Production Engineering

Protected first attempt | Retail fulfillment platform | RC1

Scenario

A retailer has a deliberately weak cloud data design. Orders arrive as daily files and operational updates. The existing design uses manual notebook runs, full reloads, broad permissions, plaintext secrets, no automated quality gate, no rollback plan and almost no monitoring. Redesign it as a production-minded Microsoft Fabric data platform and prove the local data/security controls.

Your mission

- Inspect ``weak_architecture.md`` and create a ranked risk register with at least 8 meaningful findings.
- Design a coherent Fabric target architecture using OneLake/Lakehouse, Bronze-Silver-Gold responsibilities, Spark/Delta transformation, SQL serving and managed pipeline/orchestration.
- Implement the local quality gate so invalid/duplicate records are quarantined and the expected summary is produced.
- Implement PII masking for the supplied customer email field.
- Define a watermark + merge/upsert incremental plan with rerun safety and late-data handling.
- Design least-privilege access and secret handling without plaintext production credentials.
- Create CI/CD evidence that tests code/quality/security before promotion and explains dev -> test -> prod plus rollback.
- Create monitoring/alerting and an incident runbook covering detect, diagnose, contain, recover, validate and communicate.
- Complete the provider-light Terraform foundation without hardcoded credentials.
- Explain cost/scaling choices and how the design changes at 10x data volume.
- Run ``python check_gate.py`` and save the output. Live Fabric execution evidence is collected separately before FINAL.

Protected rules

- Documentation allowed.
- No direct AI solution until the attempt folder and first checker output are saved.
- Do not open Gate 6A Review before saving the attempt.
- Do not place real keys, passwords or tokens in any file.
- A green offline checker does not replace the later live Fabric runtime pass.

Submission evidence

- Saved original attempt folder
- Checker output
- Risk register and target architecture
- Quality/masking outputs
- Security, CI/CD, monitoring, runbook and IaC evidence
- Short architecture defense